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Discrete Time Linear Time Invariant Systems

GATE ECE Signals and Systems · Previous Year Questions with Answer Key

28	21	1988 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	9 questions
Marks 2	15 questions
Marks 4	1 question
Marks 5	3 questions
Answer key	at the end

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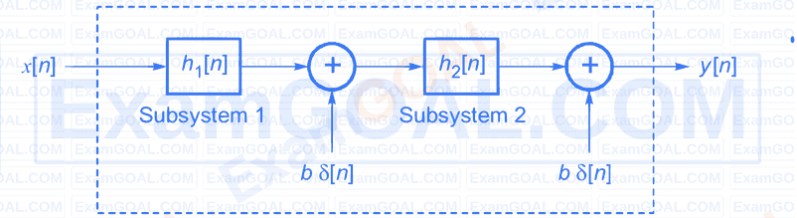
Marks 1 9 questions

Q1. GATE ECE 2025

Open solution

Consider the discrete-time system below with input $x[n]$ and output $y[n]$. In the figure, $h_1[n]$ and $h_2[n]$ denote the impulse responses of LTI Subsystems 1 and 2, respectively. Also, $\delta[n]$ is the unit impulse, and $b > 0$.

Assuming $h_2[n] \neq \delta[n]$, the overall system (denoted by the dashed box) is_____.



(A) linear and time invariant
(B) linear and time variant
(C) nonlinear and time invariant
(D) nonlinear and time variant

Q2. GATE ECE 2023

Open solution

Consider a system with input $x(t)$ and output $y(t) = x(e^t)$. The system is

(A) Causal and time invariant.
(B) Non-causal and time varying.
(C) Causal and time varying.
(D) Non-causal and time invariant.

Q3. GATE ECE 2020

Open solution

The output $y[n]$ of a discrete - time system for an input $x[n]$ is

$$y[n] = \max_{-\infty \leq k \leq n} |x[k]|$$

The unit impulse response of the system is

(A) unit step signal $u[n]$.
(B) 0 for all n.
(C) unit impulse signal $\delta[n]$.
(D) 1 for all n.

Q4. GATE ECE 2017 Set 2

Open solution

An **LTI** system with unit sample response
$h(n) = 5\delta[n] - 7\delta[n-1] + 7\delta[n-3] - 5\delta[n-4]$ is a

(A) low pass filter (B) high pass filter (C) band pass filter (D) band stop filter

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Q5. GATE ECE 2017 Set 1

Open solution

Consider a single input single output discrete-time system with $h[n]$ as input and $y[n]$ as output, where the two are related as

$$y[n] = \begin{cases} n|x[n]|, & for\ 0 \le n \le 10 \\ x[n] - x[n-1], & otherwise, \end{cases}$$

Which one of the following statements is true about the system?

(A) It is causal and stable
(B) It is causal but not stable
(C) It is not causal but stable
(D) It is neither causal nor stable

Q6. GATE ECE 2011

Open solution

A system is defined by its impulse response $h(n) = 2^n u(n-2)$. The system is

(A) stable and causal.
(B) causal but not stable.
(C) stable but not causal.
(D) unstable and non causal.

Q7. GATE ECE 2010

Open solution

Two discrete time systems with impulse responses $h_1[n] = \delta[n-1]$ and $h_2[n] = \delta[n-2]$ are connected in cascade. The overall impulse response of the cascaded system is

(A) $\delta[n-1] + \delta[n-2]$
(B) $\delta[n-4]$
(C) $\delta[n-3]$
(D) $\delta[n-1]\delta[n-2]$

Q8. GATE ECE 2004

Open solution

The impulse response $h[n]$ of a linear time-invariant system is given by $h[n] = u[n+3] + u[n-2] - 2u[n-7]$, where $u[n]$ is the unit step sequence. The above system is

(A) stable but not causal.
(B) stable and causal.
(C) causal but unstable.
(D) unstable and not causal.

Q9. GATE ECE 2003

Open solution

A sequence $x(n)$ with the z-transform $X(z) = z^4 + z^2 - 2z + 2 - 3z^{-4}$ is applied as an input to a linear, time-invariant system with the impulse response $h(n) = 2\delta(n-3)$ where $\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & otherwise \end{cases}$

The output at $n = 4$ is

(A) -6
(B) zero
(C) 2
(D) -4

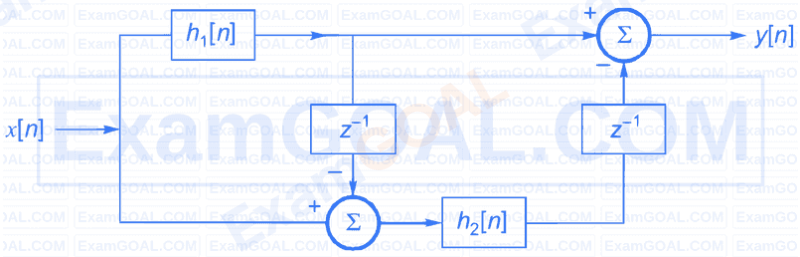
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Marks 2 15 questions

Q10. GATE ECE 2026

Consider the discrete time system (S) with input $x[n]$ and output $y[n]$ as shown in the Figure. The two sub-systems represented by their impulse responses $h_1[n]$ and $h_2[n]$ are linear and time invariant.

Which of the following statements is necessarily TRUE?



(A) S is causal.

(B) S is linear and time invariant.

(C) S is linear and time varying.

(D) S is non-linear.

Open solution

Q11. GATE ECE 2021

For a unit step input $u[n]$, a discreate-time LTI system produces an output signal $(2\delta[n+1]+\delta[n]+\delta[n-1])$. Let $y[n]$ be the output of the system for an input $\left(\left(\frac{1}{2}\right)^n u[n]\right)$.

The value of $y[0]$ is The value of $y[0]$ is ____

Open solution

Q12. GATE ECE 2015 Set 3

Two sequence $x_1[n]$ and $x_2[n]$ have the same energy.
Suppose $x_1[n] = \alpha\, 0.5^n\, u[n]$, where α is a positive real number and $u[n]$ is the unit step sequence. Assume

$$x_2[n] = \begin{cases} \sqrt{1.5} & for\ \ n = 0, 1 \\ 0 & otherwise \end{cases}$$

Then the value of α is_____.

Open solution

Q13. GATE ECE 2014 Set 2

Consider a discrete-time signal

$$x[n] = \begin{cases} n\ for\ 0 \le n \le 10 \\ 0\ otherwise \end{cases}$$

If $y[n]$ is the convolution of $x[n]$ with itself, the value of $y[4]$ is

Open solution

Q14. GATE ECE 2012

Let $y[n]$ denote the convolution of $h[n]$ and $g[n]$, where $h[n] = (1/2)^n\, u[n]$ and $g[n]$ is a causal sequence. If $y[0] = 1$ and $y[1] = 1/2$, then $g[1]$ equals

(A) 0 (B) 1/2 (C) 1 (D) 3/2

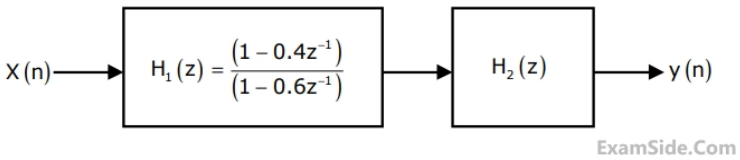
Open solution

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Q15. GATE ECE 2011

Open solution

Two system $H_1(z)$ and $H_2(z)$ are connected in cascade as shown below. The overall output $y(n)$ is the same as the input $x(n)$ with a one unit delay. The transfer function of the second system $H_2(z)$ is



(A) $\frac{(1-0.6\,z^{-1})}{z^{-1}(1-0.4\,z^{-1})}$

(B) $\frac{z^{-1}(1-0.6\,z^{-1})}{(1-0.4\,z^{-1})}$

(C) $\frac{z^{-1}(1-0.4\,z^{-1})}{(1-0.6\,z^{-1})}$

(D) $\frac{(1-0.4\,z^{-1})}{z^{-1}(1-0.6\,z^{-1})}$

Q16. GATE ECE 2010

Open solution

The transfer function of a discrete time **LTI** system is given by

$H(z) = \frac{2-\frac{3}{4}z^{-1}}{1-\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}}$

Consider the following statements:

S1: The system is stable and causal for $ROC: \ |z| > 1/2$

S2: The system is stable but not causal for $ROC: \ |z| < 1/4$

S3: The system is neither stable nor causal for $ROC: \ 1/4 < |z| < 1/2$

Which one of the following statements is valid?

(A) Both S1 and S2 are true

(B) Both S2 and S3 are true

(C) Both S1 and S3 are true

(D) S1, S2 and S3 are all true

Q17. GATE ECE 2008

Open solution

A discrete time linear shift - invariant system has an impulse response $h[n]$ with $h[0] = 1, \ h[1] = -1, \ h[2] = 2$, and zero otherwise. The system is given an input sequence $x[n]$ with $x[0] = x[2] = 1$, and zero otherwise. The number of nonzero samples in the output sequence $y[n]$, and the value of $y[2]$ are, respectively

(A) 5, 2

(B) 6, 2

(C) 6, 1

(D) 5, 3

Q18. GATE ECE 2006

Open solution

A system with input $x(n)$ and output $y(n)$ is given as $y(n) = \left(\sin \frac{5}{6}\pi n\right)x(n)$. The system is

(A) linear, stable and invertible.

(B) non-linear, stable and non-invertible.

(C) linear, stable and non-invertible.

(D) linear, unstable and invertible.

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Q19. GATE ECE 2004

A causal **LTI** system is described by the difference equation $2y[n] = ay[n-2] - 2x[n] + \beta x[n-1]$. The system is stable only if

(A) $|\alpha| = 2,\ |\beta| < 2$

(B) $|\alpha| > 2,\ |\beta| > 2$

(C) $|\alpha| < 2$, any value of β

(D) $|\beta| < 2$, any value of α

Open solution

Q20. GATE ECE 2004

The impulse response $h[n]$ of a linear time invariant system is given as

$$h[n] = \begin{cases} -2\sqrt{2}, & n = 1, -1 \\ 4\sqrt{2}, & n = 2, -2 \\ 0, & otherwise \end{cases}$$

If the input to the above system is the sequence $e^{j\pi n/4}$, then the output is

(A) $4\sqrt{2}\, e^{j\pi n/4}$

(B) $4\sqrt{2}\, e^{-j\pi n/4}$

(C) $4e^{j\pi n/4}$

(D) $-4e^{j\pi n/4}$

Open solution

Q21. GATE ECE 2003

Let P be linearity, Q be time-invariance, R be causality and S be stability.
A discrete time system has the input-output relationship,

$$y(n) = \begin{cases} x(n), & n \geq 1 \\ 0, & n = 0 \\ x(n+1), & n \leq -1 \end{cases}$$

Where $x(n)$ is the input and $y(n)$ is the output. The above system has the properties

(A) P, S but not Q, R

(B) P, Q, S but not R

(C) P, Q, R, S

(D) Q, R, S but not P

Open solution

Q22. GATE ECE 2002

If the impulse response of a discrete-time system is $h[n] = -5^n\, u[-n-1]$, then the system function $H(z)$ is equal to

(A) $\frac{-z}{z-5}$ and the system is stable.

(B) $\frac{z}{z-5}$ and the system is stable.

(C) $\frac{-z}{z-5}$ and the system is unstable.

(D) $\frac{z}{z-5}$ and the system is unstable.

Open solution

Q23. GATE ECE 1992

A linear discrete - time system has the characteristic equation, $z^3 - 0.81\ z = 0$. The system

(A) is stable.

(B) is marginally stable.

(C) is unstable.

(D) stability cannot be assessed from the given information.

Open solution

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Q24. GATE ECE 1988

Open solution

Consider the system shown in the Fig.1 below. The transfer function $Y(z)/X(z)$ of the system is

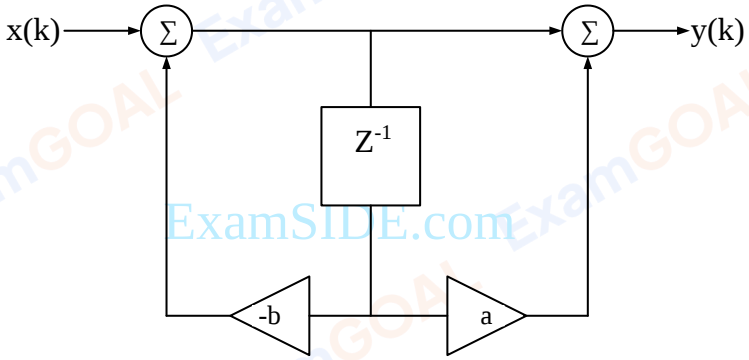


Fig.1

(A) $\frac{1+a\,z^{-1}}{1+b\,z^{-1}}$

(B) $\frac{1+b\,z^{-1}}{1+a\,z^{-1}}$

(C) $\frac{1+a\,z^{-1}}{1-b\,z^{-1}}$

(D) $\frac{1-b\,z^{-1}}{1+a\,z^{-1}}$

Marks 4 1 questions

Q25. GATE ECE 1988

Open solution

The output of a system is given in difference equation form as $y(k) = a\,y(k-1) + x(k)$, where $x(k)$ is the input. If $x(k) = 0$ for $k \neq 0$, $x(0) = 1$, and $y(0) = 0$, find $y(k)$ for all k. Determine the range of $'a'$ for which $y(k)$ is bounded.

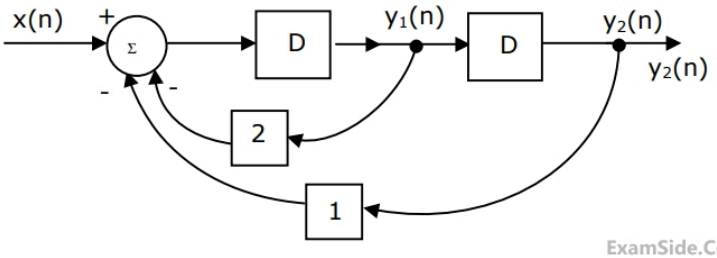
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Marks 5 3 questions

Q26. GATE ECE 1997

In Fig. 1, a linear time invariant discrete system is shown. Blocks labeled D represent unit delay elements. For $n < 0$, you may assume that $x(n), y_1(n), y_2(n)$ are all zero.

Open solution



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Fig.21

(a) Find the expression for $y_1(n)$ and $y_2(n)$ in terms of $x(n)$.
(b) Find the transfer function $y_2(z)/X(z)$ in the z-domain.
(c) If $x(n) = 1$ at $n = 0$ or $x(n) = 0$ otherwise

Find $y_2(n)$.

Q27. GATE ECE 1996

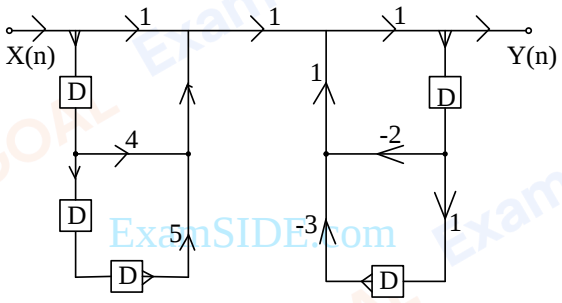
A system having a unit impulse response $h(n) = u(n)$ is excited by a signal $x(n) = \alpha^n u(n)$. Determine the output $y(n)$

Open solution

Q28. GATE ECE 1996

In the linear time-invariant system shown in Fig. 1, blocks labeled D represent unit delay elements. Find the expression for $y(n)$, and also the transfer function $\frac{Y(z)}{X(z)}$ in the z-domain.

Open solution



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Fig. 1

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Answer Key

Marks 1

1. **D**	2. **B**	3. **A**	4. **C**	5. **A**
6. **B**	7. **C**	8. **A**	9. **B**	

Marks 2

10. **B**	11. **0**	12. **1.5**	13. **10**	14. **A**
15. **B**	16. **C**	17. **D**	18. **C**	19. **C**
20. **D**	21. **A**	22. **B**	23. **A**	24. **A**

Marks 4

25. $y(k)$ **is bounded where** $\lvert a\rvert > 1$**.**				

Marks 5

26. **(a)** $y_1(n) = x(n-1) - 2y_1(n-1) - y_1(n-2)$ $y_2(n) = x(n-2) - 2y_2(n-1) - y_2(n-2)$ **(b)** $\frac{y_2(z)}{x(z)} = \frac{1}{(z+1)^2}$ **(c)** $y_2(n) = 0$ **where** $n \leq 0$	27. $y(n) = 0$, **where** $n \leq -1$	28. $\frac{Y(z)}{X(z)}$ $\frac{Y(z)}{X(z)} = \frac{z^3+4z^2+5}{z(z^2+2z+3)}$		

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